Reg. No. | | | | | | | | | | |

[22MSCH3H04]

THIRD SEMESTER UNIVERSITY EXAMINATION, NOVEMBER 2023

ORGANIC CHEMISTRY–III

Time: 03 HOURS **Maximum Marks: 50**

SECTION – A

1. Answer any FOUR of the following questions. **4×5=20**

a. What is functional group interconversion (FGI)? Give an example. Briefly discuss one group and two group C-X disconnection with suitable examples.

b. How one can use Gilman reagent to introduce two different alkyl groups in one enol?

c. Geometry of ketenes allow a suprafacial [2+2] cycloaddition with alkenes in presence of photons, unlike in thermal conditions. Explain it by molecular orbital correlation.

d. Describe the mechanism involved in the following reaction.

e. Briefly explain mutarotation with a suitable example. What are anomers? Give an example.

SECTION – B

2. Answer any TWO of the following questions. **2×9=18**

a. Elucidate acyclic stereoselection in 1, 2 and 1, 3- asymmetric induction.

b. Explain the mechanism of Peterson synthesis with an example.

c. Discuss the properties of monosaccharides in detail.

SECTION – C

3. Answer the following question. **12**

a. **i.** Show the retrosynthetic analysis and synthetic method for saccharine.

ii. Write note on stereochemistry of electrocyclic reaction. Explain the terms conrotatory and disrotatory involved in it.

iii. How to convert pyrimidine to pyrazole. Explain with mechanism.